

Casper Versteeg · U.S. Citizen

Experienced and adept engineering leader with an extensive spacecraft design and analysis record. Successfully applied a systems-minded approach in startup environments to build strong teams and deliver high-impact technologies for customers.

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EXPERIENCE

• Turion Space Corp.

Irvine, CA

Director of Advanced Projects

October 2025 - August 2026

- **Leadership:** Spearhead cutting-edge research initiatives in satellite subsystem technologies, translating emerging concepts into viable hardware solutions for spacecraft. Oversee a team of engineers and researchers, fostering innovation and collaboration to drive project success.
- **Vertical Integration Strategy:** Architected and executed in-house development of ADCS components including reaction wheels, magnetorquers and star trackers, reducing supply chain dependencies and reducing unit cost by 90%.
- **System-Level Analysis:** Conduct comprehensive trade studies and mission architecture assessments for prospective programs, evaluating technical feasibility, cost implications, and risk factors to inform strategic decision-making.
- **Modeling and Simulation:** Modeled conceptual space missions using both in-house software tools and custom-developed analysis code to evaluate architecture tradeoffs, performance, and feasibility. Used results to inform mission design and system budgets.
- **Cross-Functional Leadership:** Bridge technical and commercial operations between Business Development teams to shape customer proposals and capture strategies while coordinating with engineering teams across mechanical, electrical and software disciplines to ensure integrated system solutions meet mission and customer requirements.

Director of Analysis and Systems Engineering

March 2025 - October 2025

- **Leadership:** Director for analysis and systems engineering teams, coordinate with respective team leads to ensure timely and accurate analysis and design work.
- **Systems Engineering:** Lead systems engineering efforts for multiple projects, including spacecraft design, payload design, and launch vehicle integration. Responsible for ensuring that all systems engineering processes are followed and that all documentation is complete and accurate.
- **Design Reviews:** Successfully led systems engineering and analysis efforts for \$50M+ contracts with USSF/AFRL, resulting in successful SRR reviews.
- **Proposals:** Wrote proposals and technical volumes for business capture efforts with potential value of \$1.5B.

Lead Vehicle Engineer

May 2023 - March 2025

- **Leadership:** Managed four cross-functional teams consisting of 20 engineers across mechanical design, analysis, GNC, and systems engineering; improving communication between siloed teams. Wrote proposals and technical volumes for projects funded by USSF/AFRL, resulting in \$2M+ of funding awarded.
- **Analysis:** Lead analysis efforts of design projects, including the primary spacecraft and payloads, using NASTRAN, ThermalDesktop, and custom analysis tools made in Python.
- **Testing:** Designed thermal and structural tests to characterize flight hardware. Presently involved in designing test hardware to demonstrate a reversible dry-adhesive capture mechanism.
- **Documentation:** Documented analysis and design work, culminating in presentations for PDR and CDR. Prepared reports for documenting work in technical efforts performed for different projects.

- Lynk Global, Inc.** Fall Church, VA
Testing and Analysis Lead *April 2021 - May 2023*
 - **Analysis:** Responsible for all structural and thermal analysis of spacecraft designs from component- to space vehicle level, using NASTRAN (structural), TMG (thermal), as well as my own codes. Prepared CLA models for launch provider.
 - **Design:** Developed parameterized models and scripts to optimize spacecraft structures for stiffness-to-mass ratio, using design equations and numerical analysis; interfaced with systems, software, and electrical engineering teams to inform payload mechanical design and layout.
 - **Testing:** Streamlined testing operations through objective-oriented application of the scientific method; planned and designed procedures for environmental testing of all payload hardware; developed test-analysis models for model correlation to test data.
 - **Documentation:** Prepared technical reports documenting analysis/tests performed in support of payload/spacecraft design, development, launch, and operations.
- Dolbow Research Group** Durham, NC
Research Assistant *August 2018 - May 2021*
 - **Analysis:** Trained as engineering analyst, specializing in mathematical and computational modeling.
 - **Dynamic Fracture:** Developed and studied a regularized fracture model, based on phase-field theory, for fast, brittle fracture. Implemented model in existing parallelized research code.
 - **Fluid-Structure Interaction:** Modeled interaction of high Reynolds number flow with flexible structures, with applications to flow-induced vibration.
- Small Satellite Research Lab** Athens, GA
Mechanical Team Lead *January 2017 - August 2018*
 - **Leadership:** Supervised the mechanical team; onboarded and trained new members, directed team tasks, managed deadlines; approved mechanical designs for manufacturing.
 - **Systems Engineering:** Worked closely with interdisciplinary teams, tackling mechanical, electrical and software design challenges for a holistic approach to ensuring mission success.
 - **Design/Analysis:** Designed science payloads for two CubeSat missions; validated designs through structural and thermal models of spacecraft components and assemblies.

SKILLS

- **Programming:** Python, MATLAB, C/C++, Git
- **Software:** Simcenter 3D, Femap, NASTRAN, ThermalDesktop, SolidWorks, AutoCAD, Inventor, TMG, ANSYS Workbench, GNU Linux, MacOS, Microsoft Windows, Microsoft Office Suite

EDUCATION

- Duke University** Durham, NC
Master of Science in Mechanical Engineering *Aug. 2018 – May 2021*
Emphasis in computational mechanics and finite element analysis
Relevant coursework: Nonlinear Finite Element Analysis, Acoustics, Continuum Mechanics, Dynamics
- University of Georgia** Athens, GA
Bachelor of Science in Mechanical Engineering; GPA: 3.89 *Aug. 2014 – May 2018*